LAB 9

Task 1

```
% AND truth table
and(0, 0, 0).
and(0, 1, 0).
and(1, 0, 0).
and(1, 1, 1).

% OR truth table
or(0, 0, 0).
or(0, 1, 1).
or(1, 0, 1).
or(1, 1, 1).

% NOT truth table
not(0, 1, 1).
not(1, 0, 1).

% Boolean expression: (A AND B) OR (NOT A AND NOT B)
expression(A, B, Result) :-
    and(A, B, X),
    not(A, NA, 1),
    not(B, NB, 1),
    and(NA, NB, Y),
    or(X, Y, Result).
```

```
93 ?- expression(0, 0, R).
R = 1.

94 ?- expression(0, 1, R).
R = 0.

95 ?- expression(1, 0, R).
R = 0.

96 ?- expression(1, 1, R).
R = 1.
```

Task 2

```
% Empty list
sumlist([], 0).

% Recursive case
sumlist([H|T], Sum) :-
    sumlist(T, Rest),
    Sum is H + Rest.
```

```
101 ?- consult(['c:/Users/alhan/OneDrive/سطح المكتب/sumlist.pl']).
true.

102 ?- sumlist([1,2,5,7], Sum).
Sum = 15.

103 ?- sumlist([], Sum).
Sum = 0.

104 ?-
```

Task 3

```
total_average :-
    write('Enter first number: '),
    read(A),
    write('Enter second number: '),
    read(B),
    write('Enter third number: '),
    read(C),
    Total is A + B + C,
    Average is Total / 3,
    write('Total = '),
    write(Total), nl,
    write('Average = '),
    write(Average), nl.
```

```
103 ?- total_average.
Enter first number: 5.
Enter second number: |: 7.
Enter third number: |: 9.
Total = 21
Average = 7
true.

103 ?- total_average.
Enter first number: 2.
Enter second number: |: 4.
Enter third number: |: 7.
Total = 13
Average = 4.333333333333333
true.
```